HARIRAMAKRRISHNAN RAMACHANDRAN

+353 89 970 6156 • hariramakrrish@gmail.com • Dublin, Ireland • **Stamp 1G — Full-Time Work Eligible**

PROFESSIONAL EXPERIENCE

Software Engineer — HCL Technologies, Chennai, India Sep 2021 – Jan 2025

- Developed and trained deep learning networks using **Python** and **scikit-learn** for object classification tasks on automotive datasets, achieving approximately **0.92 AUC** on held-out validation data.
- Selected and curated appropriate datasets, implementing robust data representation methods for training complex neural networks, ensuring data integrity for downstream analysis.
- Collaborated with data engineers to establish and maintain efficient data pipelines, processing large volumes of image and sensor data for deep learning model consumption.
- Performed statistical analysis on model performance metrics, fine-tuning network architectures and hyperparameters to optimize results based on test outcomes.
- Contributed to the development of state-of-the-art **deep learning algorithms** for scene understanding, directly supporting the creation of next-generation automated driving platforms.
- Maintained an up-to-date understanding of technological advancements in **deep learning** and their practical applications within the automotive sector, sharing insights with the team.
- Containerized deep learning training environments using **Docker** and managed deployments on **AWS EC2** instances, enabling reproducible research and development cycles.

SKILLS

Machine Learning – Deep Learning, Neural Networks, scikit-learn, Keras, PyTorch, Classification, Statistical Analysis

Programming Languages – Python, English

Computer Vision – OpenCV, 3D Object Detection, Pedestrian Classification, Trajectory Mapping, Sensor Fusion

Data Engineering – Data Pipelines, Data Integrity, Data Acquisition, Data Representation

Software Development – Linux, Git, Software Architecture, Data Structures, Data Modeling

Cloud & DevOps – AWS, Docker, CI/CD

PROJECTS

Deep Learning for Automotive Object Detection

- Developed a convolutional neural network (CNN) using **PyTorch** and **OpenCV** to detect and classify vehicles and pedestrians in simulated driving environments.
- Trained the model on a custom dataset of 150,000 annotated images, achieving a **94% precision** for vehicle detection and **89% recall** for pedestrian detection.
- Implemented data augmentation techniques, including random cropping and color jittering, to improve model generalization and robustness against varying environmental conditions.

Automated Lane Detection System

- Engineered a Python-based system utilizing traditional computer vision algorithms and **scikit-learn** for identifying lane markings from front-facing camera feeds.
- Processed real-time video streams, extracting features and applying filtering methods to reliably detect lane boundaries under diverse lighting and road surface conditions.
- Integrated the lane detection module with a simulated vehicle control interface, demonstrating its potential for driver-assistance functionalities.

EDUCATION

M.Sc. Data Analytics — National College of Ireland, Dublin Feb 2026

B.E. Computer Science — SNS College of Technology, Coimbatore, India 2021

CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)